Space Vector Modulation v4.2 User Guide

Introduction (Ask a Question)

Space Vector Modulation (SVM) is an improved technique for generating a fundamental sine wave that provides a higher voltage to the motor in a 3-phase system, lowers total harmonic distortion, and controls the number of short pulses in the Pulse Width Modulation (PWM) waveform.

Summary (Ask a Question)

Core Version	This document applies to Space Vector Modulation v4.2.
Supported Device Families	• PolarFire® SoC • PolarFire • RTG4™ • IGLOO® 2 • SmartFusion® 2
Supported Tool Flow	Requires Libero® SoC v11.8 or later releases.
Licensing	Complete encrypted RTL code is provided for the core, enabling the core to be instantiated with SmartDesign. Simulation, Synthesis, and Layout can be performed with Libero software. SVM is licensed with encrypted RTL that must be purchased separately. For more information, see SVM.

Features (Ask a Question)

SVM has the following key features:

- The Min-Max method is implemented
- The third harmonic voltage is generated from the min. and max. values of instantaneous 3-phase reference voltages
- The third hormonic phase voltages are computed

Implementation of IP Core in Libero Design Suite (Ask a Question)

IP core must be installed to the IP Catalog of the Libero® SoC software. This is installed automatically through the IP Catalog update function in the Libero SoC software, or the IP core can be manually downloaded from the catalog. Once the IP core is installed in the Libero SoC software IP Catalog, the core can be configured, generated, and instantiated within the SmartDesign tool for inclusion in the Libero project list.

Device Utilization and Performance (Ask a Question)

The following table lists the device utilization used for SVM.

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Table 1. SVM Utilization

Device Details		Resources		Performance (MHz)	RAMs		Math Blocks	Chip Globals
Family	Device	LUTs	DFF		LSRAM	µSRAM		
PolarFire® SoC	MPFS250T	207	93	200	0	0	0	0
PolarFire	MPF300T	207	93	200	0	0	0	0
SmartFusion® 2	M2S150	207	93	200	0	0	0	0

Important:

1. The data in this table is captured using typical synthesis and layout settings. CDR reference clock source was set to **Dedicated** with other configurator values unchanged.
2. Clock is constrained to 200 MHz while running the timing analysis to achieve the performance numbers.

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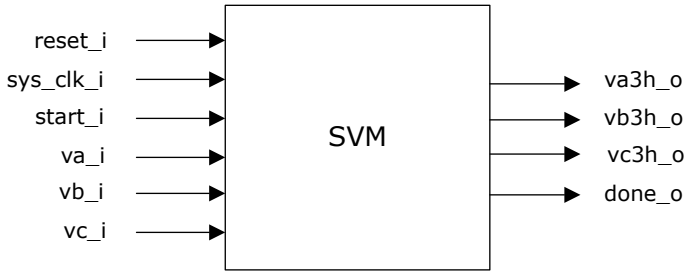
1. Functional Description (Ask a Question)

This section describes the implementation details of the SVM.

The following figure shows the system-level block diagram of the SVM.

Figure 1-1. System-Level Block Diagram of SVM



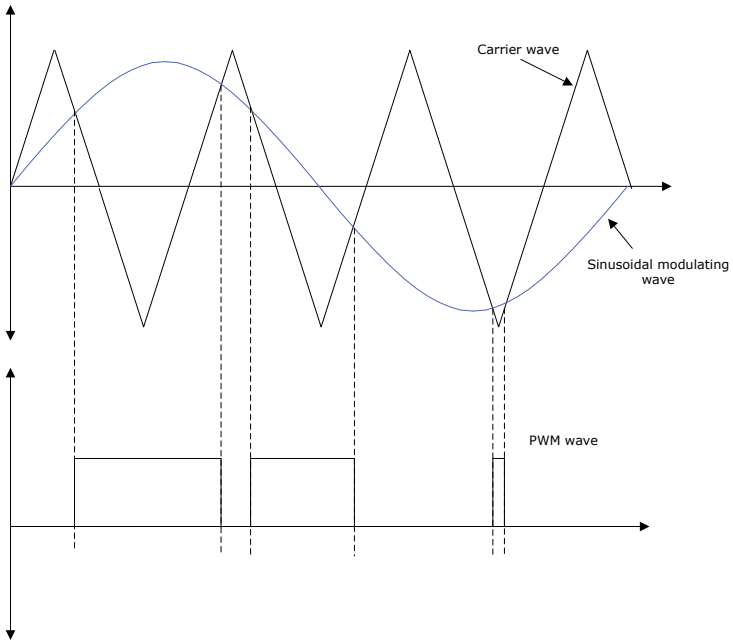
The SVM block is a control unit which implements the Min-Max method discussed in the Min-Max Method. The computations involved in the Min-Max method are performed within the block.

The entire system is synchronized with a system clock given to sys_clk_i and controlled by a Finite State Machine (FSM) and then implemented in the SVM block.

1.1 Sinusoidal PWM (Ask a Question)

A sinusoidal PWM drive cannot provide a line-line output voltage as high as the line supply.

Figure 1-2. Sinusoidal PWM Waveform



SVM is a technique that is used to increase the output voltage of PWM drive and reduce the number of short pulses. The strategy used in SVM technique is to modify the modulation of sine waves (input phase voltages) to increase the inverter voltage gain and also to reduce inverter losses.

1.2 Min-Max Method (Ask a Question)

There are many techniques to implement SVM out of which Min-Max method requires least computation and is used in the IP block. The third harmonic voltage is generated from min. and max. values of instantaneous 3-phase reference voltages. The third harmonic voltage (V_{3H}) is calculated as:

$$V_{3H} = \frac{\max(V_a, V_b, V_c) + \min(V_a, V_b, V_c)}{2}$$

The computed third harmonic voltage is added to each phase voltage to get space vector modulated reference.

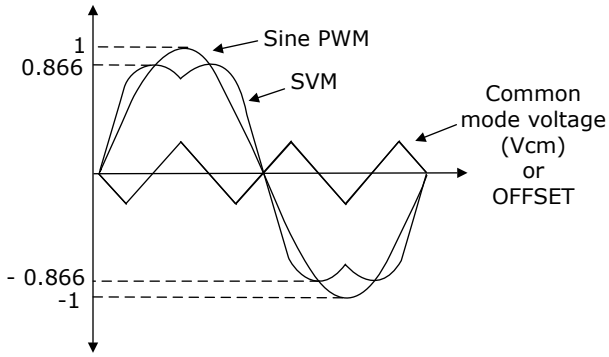
$$V_{SA} = V_A - V_{3H}$$

$$V_{SB} = V_B - V_{3H}$$

$$V_{SC} = V_C - V_{3H}$$

The addition of third harmonic reduces the peak of the reference signal by 15% and hence allows more fundamental without reaching over-modulation. Even though third harmonic is added in phase voltages, it does not appear in line voltages because it is common mode voltage. As a result, there is no third harmonic current but only fundamental current. For a given modulation index, the peak modulation voltage is farther from the peak carrier voltage. Thus, the number of short pulses is minimized. To utilize the DC bus fully, the voltages must be scaled up by 15% in one of the blocks that follows the SVM in the FOC loop.

Figure 1-3. SVM Min-Max Method



If the output should be a unity peak value and the reduction in peak introduced by the block is 0.866, the factor k must be 1.155 (k * 0.866 = 1, k=1/0.866).This implies that there is a 15.5 percent boost introduced in the phase voltage, which results in better utilization of the DC bus.

2. SVM Parameters and Interface Signals (Ask a Question)

This section discusses the parameters in the SVM GUI configurator and I/O signals.

2.1 Inputs and Outputs Signals (Ask a Question)

The following table lists the input and output ports of SVM.

Table 2-1. Inputs and Outputs of SVM

Signal Name	Direction	Description
sys_clk_i	Input	System clock
reset_i	Input	Active Low asynchronous reset signal to design
start_i	Input	A single bit start signal that must go high for one clock cycle to start SVM computations
va_i	Input	Input phase voltage (va)
vb_i	Input	Input phase voltage (vb)
vc_i	Input	Input phase voltage (vc)
va3h_o	Output	Output voltage signal corresponding to input phase voltage va_i, with the third harmonic component
vb3h_o	Output	Output voltage signal corresponding to input phase voltage vb_i, with the third harmonic component
vc3h_o	Output	Output voltage signal corresponding to input phase voltage vc_i, with the third harmonic component
done_o	Output	A single bit signal which goes High for one clock cycle to indicate that all the computations are done and output corresponding to given inputs is obtained

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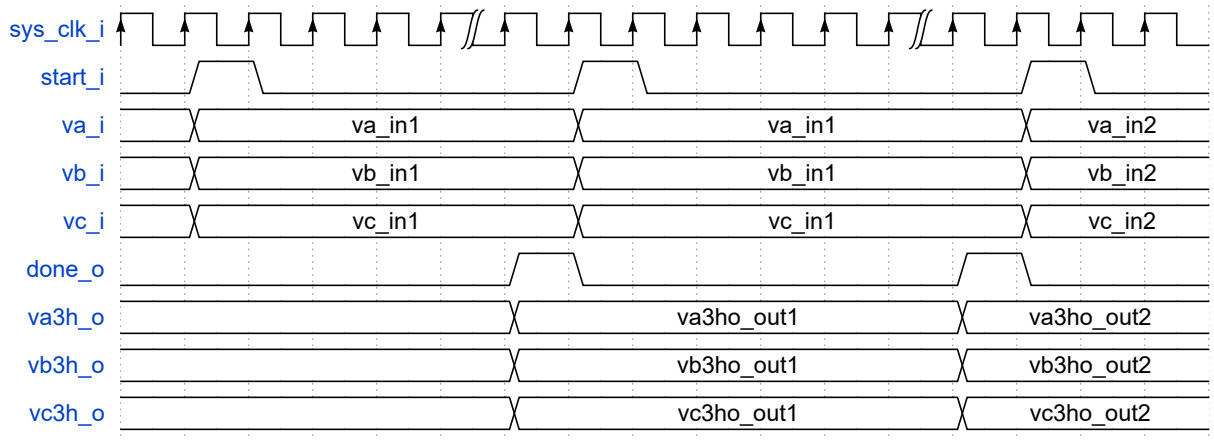
3. Timing Diagrams (Ask a Question)

This section discusses SVM timing diagrams.

The following figure shows the timing diagram of SVM.

Figure 3-1. SVM Timing Diagram



4. Testbench (Ask a Question)

A unified testbench is used to verify and test SVM called as user testbench. Testbench is provided to check the functionality of the SVM IP.

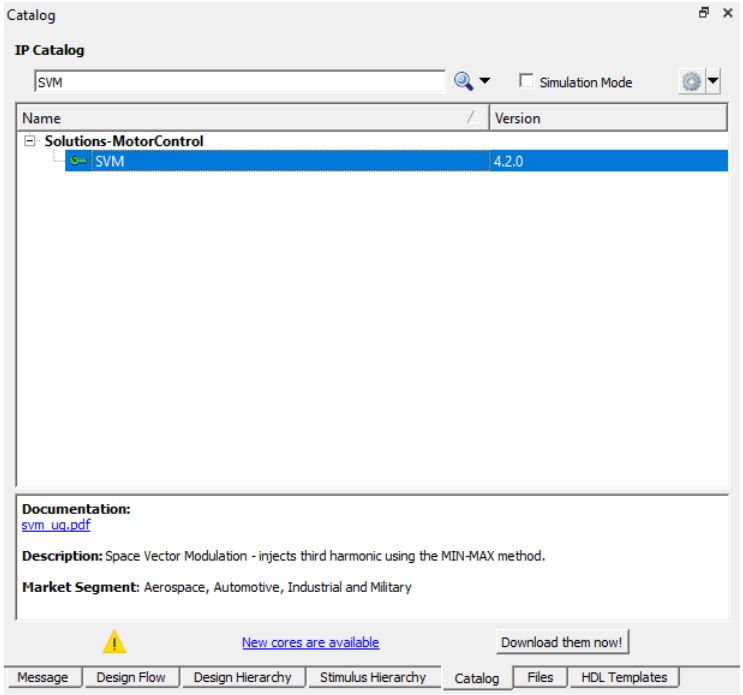
4.1 Simulation (Ask a Question)

The following steps describe how to simulate the core using the testbench:

1. Open Libero SoC **Catalog** tab, expand **Solutions-MotorControl**, double click **SVM**, and then click **OK**. The documentation associated with the IP are listed under **Documentation**.

> **Important:** If you do not see the **Catalog** tab, navigate to **View** > **Windows** menu and click **Catalog** to make it visible.

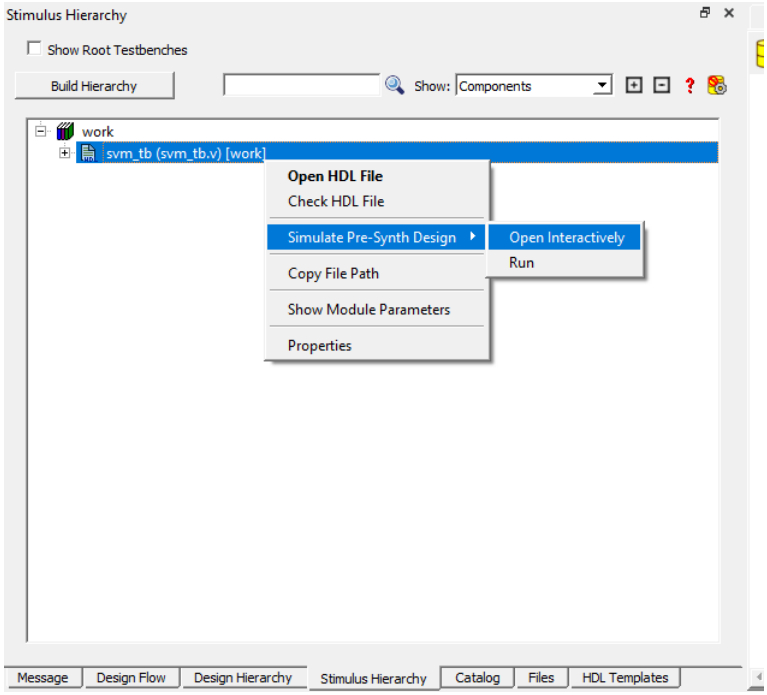
Figure 4-1. Space Vector Modulation IP Core in Libero SoC Catalog



2. On the **Stimulus Hierarchy** tab, select the testbench (`svm_tb.v`), right click and then click **Simulate Pre-Synth Design** > **Open Interactively**.

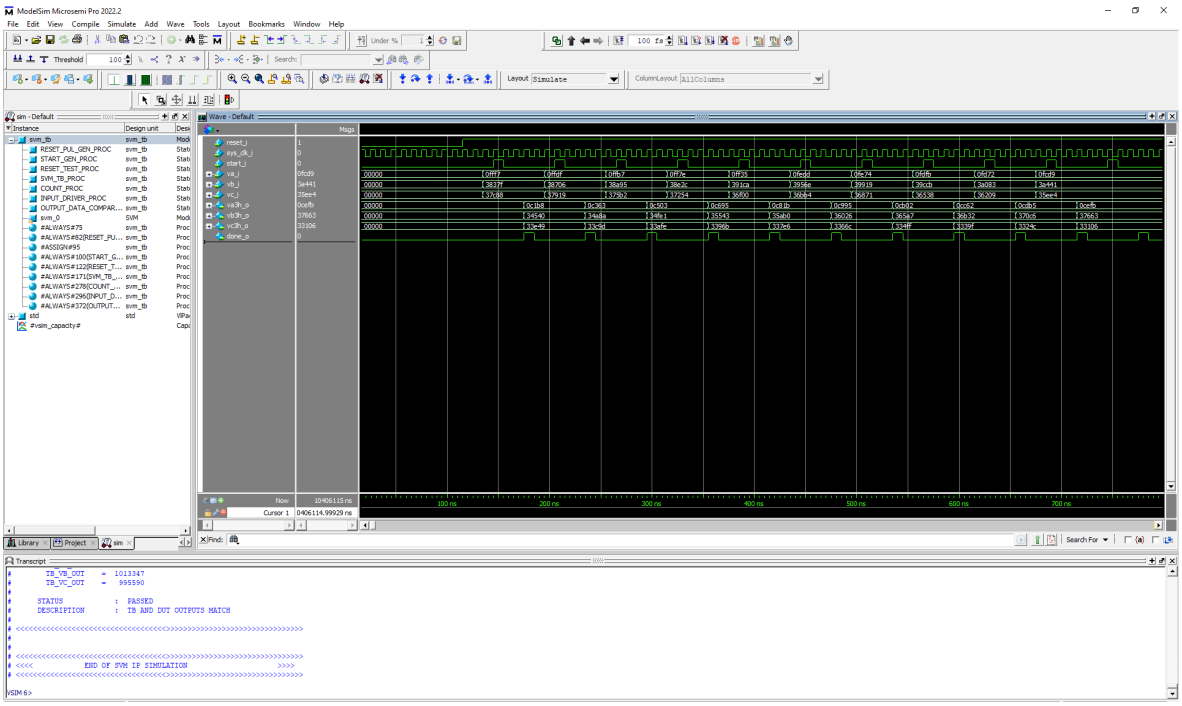
> **Important:** If you do not see the **Stimulus Hierarchy** tab, navigate to **View** > **Windows** menu and click **Stimulus Hierarchy** to make it visible.

Figure 4-2. Simulating Pre-Synthesis Design



ModelSim opens with the testbench file, as shown in the following figure.

Figure 4-3. ModelSim Simulation Window



Important: If the simulation is interrupted due to the runtime limit specified in the `.do` file, use the `run -all` command to complete the simulation.

5. Revision History (Ask a Question)

The revision history describes the changes that were implemented in the document. The changes are listed by revision, starting with the most current publication.

Table 5-1. Revision History

Revision	Date	Description
A	03/2023	The following list of changes are made in revision A of the document: • Migrated the document to the Microchip template. • Updated the document number to DS50003505A from 50200468. • Added 3. Timing Diagrams. • Added 4. Testbench.
5.0	—	The following is a summary of the changes in revision 5.0 of this document. • Added the IP version to the document title. • Removed Configuration Parameter section from Hardware Implementation.
4.0	—	Removed FSM related information from SVM Hardware Implementation and Configuration Parameters sections.
3.0	—	Changed the block name from SVPWM to SVM throughout the document.
2.0	—	The following is a summary of the changes in revision 2.0 of this document. • Updated the title of the user guide. • Updated Min-Max Method section and removed Direct Injection of Third Harmonic section. • Updated Table 1 and Table 2. • Removed Appendix.
1.0	—	Revision 1.0 was the first publication of this document.

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